The empirical recurrence for OEIS A303425: recovering a conjecture that is not written down

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Abstract

OEIS A303425 records "Empirical recurrence of order 44", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 68$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 44, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 44 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A303425 is "Number of 6 X n 0..1 arrays with every element equal to 0, 1, 2, 3, 4 or 5 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.". It has offset 1 and begins

$$32, 2048, 123732, 7486369, 454381369, 27575294129, 1673186560760, 101524906002993, \ldots$$

The entry states:

> Empirical recurrence of order 44 (see link above).

As of the "Last modified" line on the live entry (Jan 20 2024 17:08:49, revision 6) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4096$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 68$ of the 4096, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 68$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 136$ exact terms of the model returns order 44 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{44} c_i\, a(n-i),$$

c_1	43	c_{12}	145743661800	c_{23}	−9141815312842112	c_{34}	9640782328996020224
c_2	704	c_{13}	−703151746258	c_{24}	−76018064294186144	c_{35}	−13437233655547723776
c_3	17683	c_{14}	−4008050910892	c_{25}	201549094304858880	c_{36}	9365678032038789120
c_4	255614	c_{15}	−26631679894682	c_{26}	470168210237148096	c_{37}	−9680907394059337728
c_5	1640489	c_{16}	−25192525638848	c_{27}	−305576475656115328	c_{38}	−3760460111024750592
c_6	7612518	c_{17}	195893310816980	c_{28}	663122950141541376	c_{39}	9007605640184463360
c_7	−44283107	c_{18}	374324983814780	c_{29}	−1337867140021232640	c_{40}	−4367220114701942784
c_8	−1282025263	c_{19}	1147530778113704	c_{30}	−3690584397891403776	c_{41}	3373546844624781312
c_9	−5260631235	c_{20}	2342310829313512	c_{31}	3979624204007211008	c_{42}	149746131255951 36
c_{10}	24141852952	c_{21}	−10791679160757536	c_{32}	−3205683052381007872	c_{43}	−1609954500318068736
c_{11}	182713441848	c_{22}	−19642210910609792	c_{33}	4404015328650903552	c_{44}	412732527350906880

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 44, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $44+4$ terms removed returns order 44 as well, so $\deg q_{\min} = 44 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $44+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 44 that this sequence satisfies — and that family turns out to have exactly one member.

References

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